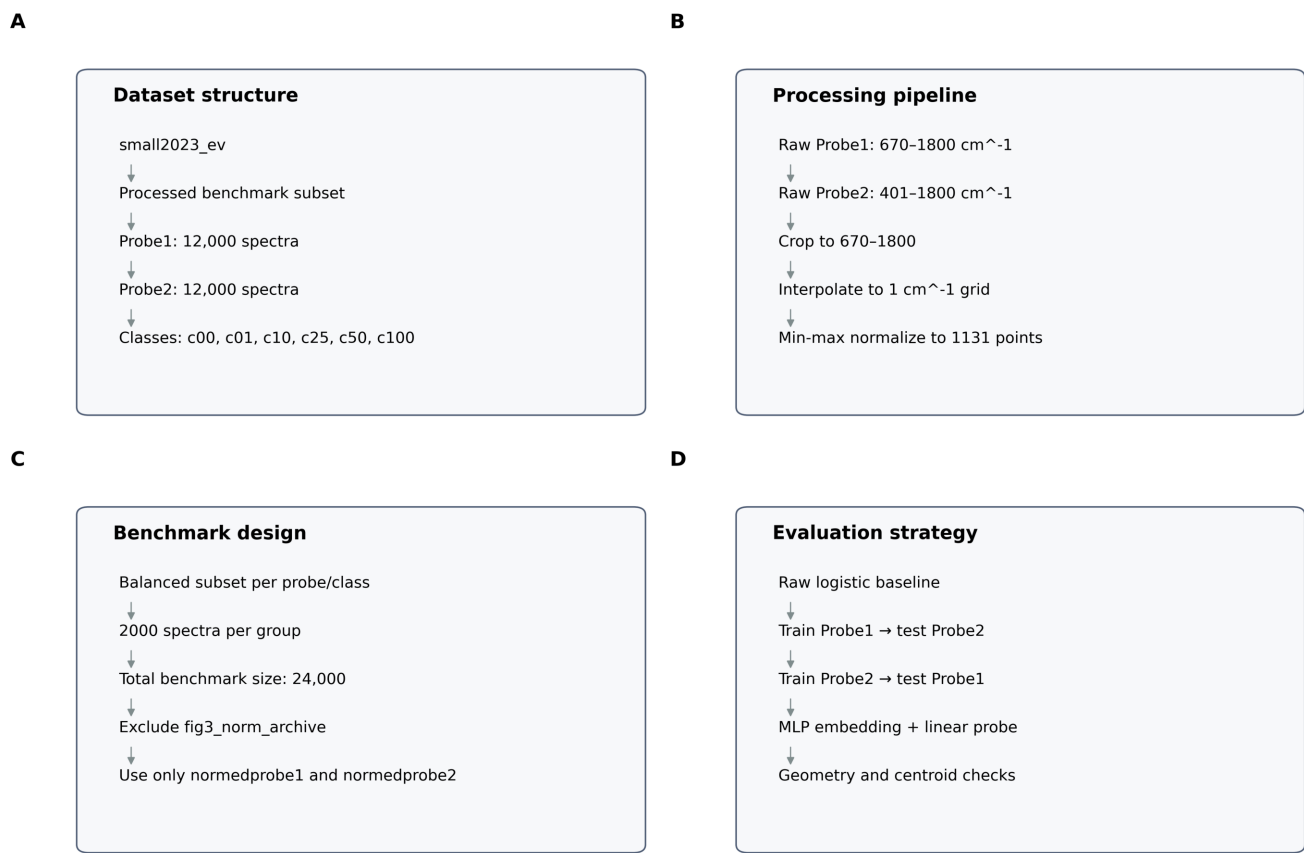


Substrate-invariant spectral embedding benchmark for small EV SERS dataset

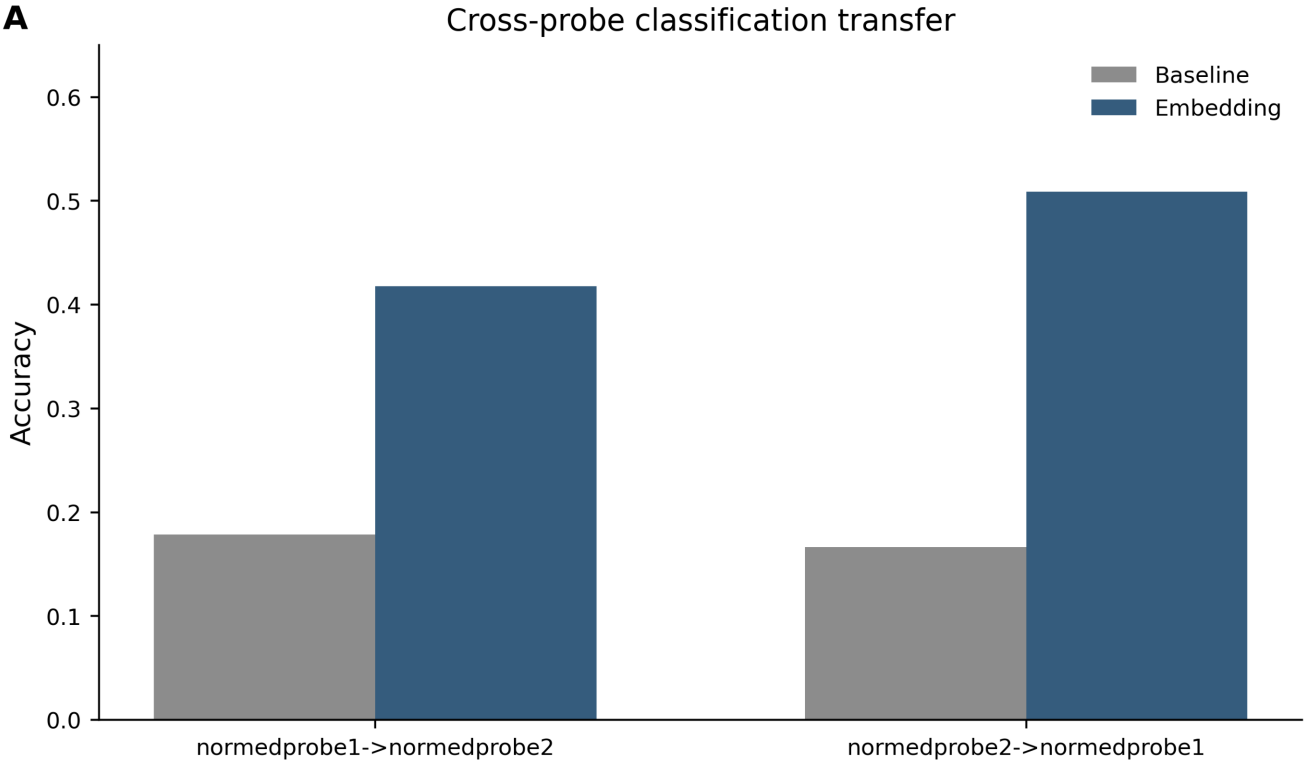
This report summarizes the completed v1 invariant embedding benchmark for small2023_ev. The analysis compares a raw processed-spectrum logistic baseline against a compact learned embedding using only normedprobe1 and normedprobe2 mixture-class spectra on the common 670-1800 cm^{-1} grid.

Figure 1. Method overview



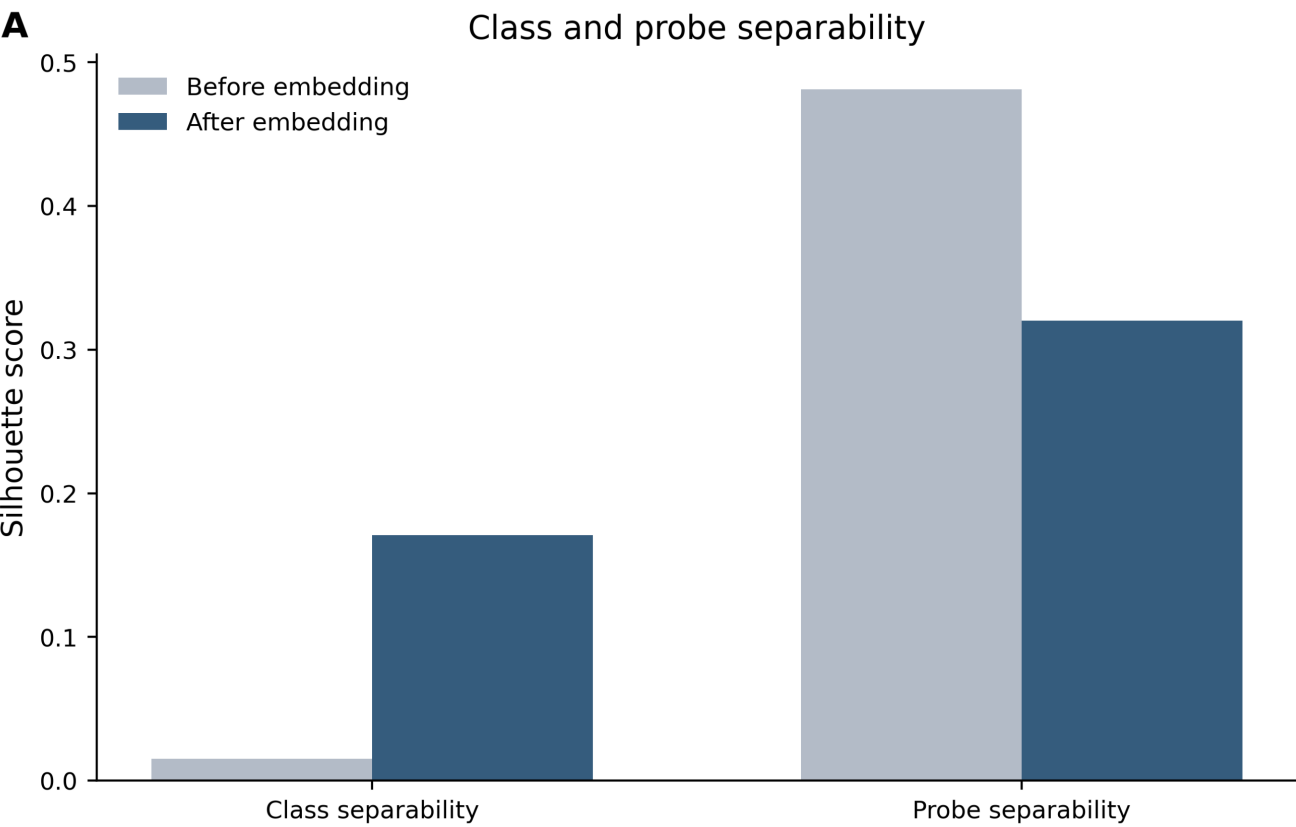
Dataset structure, processing pipeline, benchmark design, and evaluation strategy.

Figure 2. Cross-probe classification



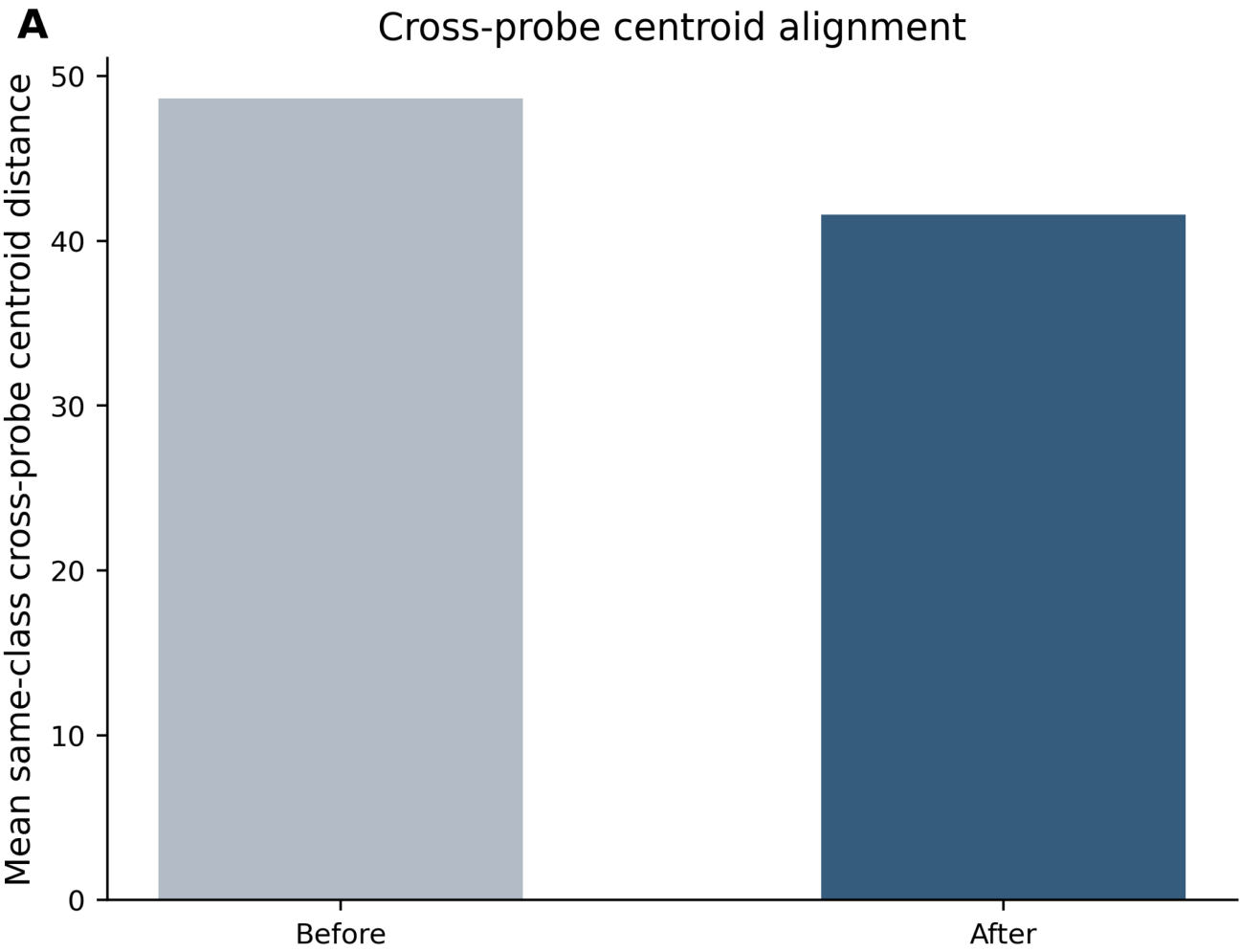
Embedding transfer improves from 0.173 mean accuracy to 0.463.

Figure 3. Separability comparison



Class silhouette improves from 0.015 to 0.171, while probe silhouette drops from 0.481 to 0.320.

Figure 4. Cross-probe centroid shift



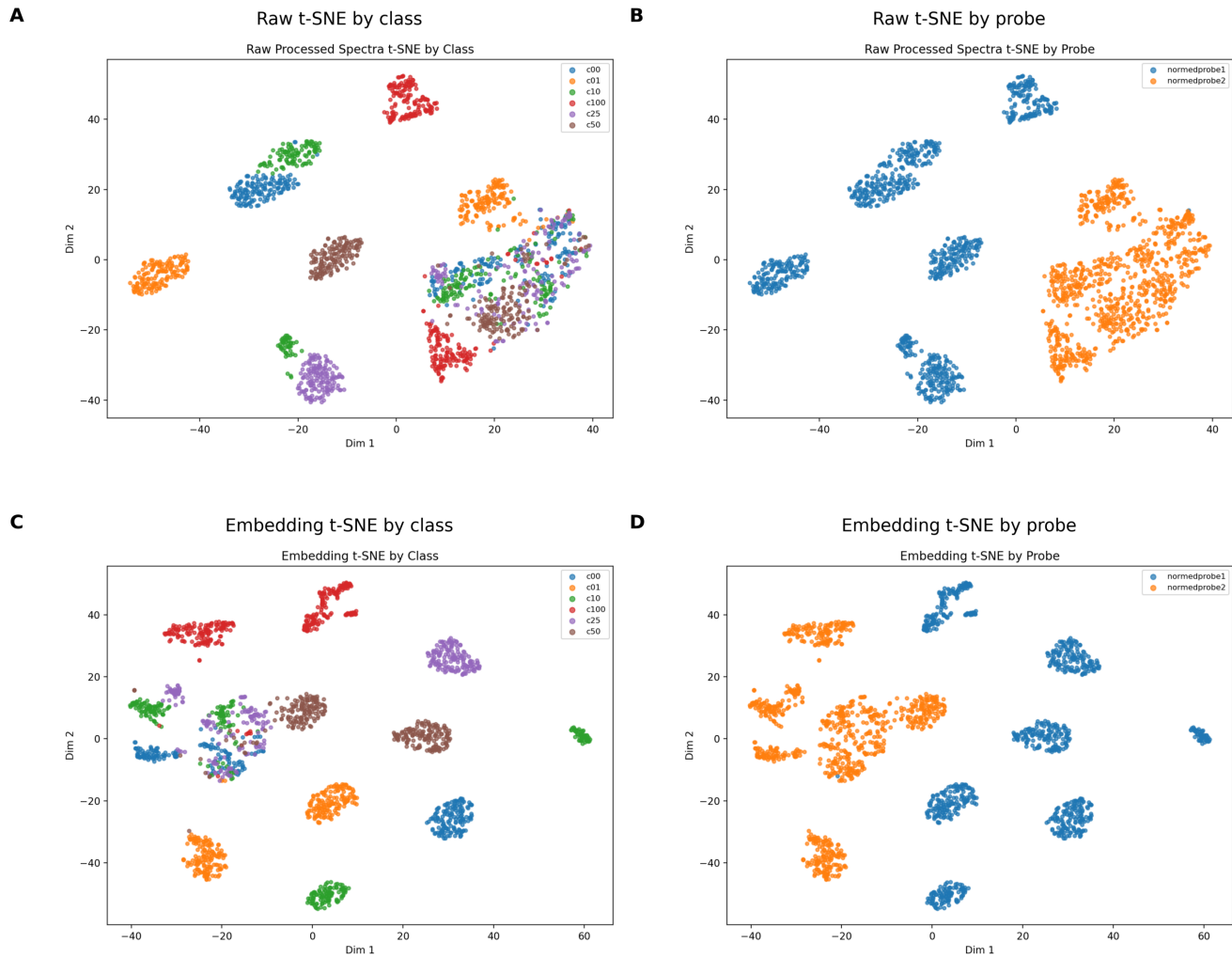
Mean same-class cross-probe centroid distance contracts from 48.653 to 41.584.

Figure 5. Mixture ordering



Mixture ordering correlation changes from 0.564 to 0.451.

Figure 6. Spectral geometry



Raw and embedded geometry colored by class and by probe.

Interpretation and limitations

The learned embedding is promising as a GAIRA benchmark because it improves cross-probe transfer, increases class separability, reduces probe separability, and moves same-class probe centroids closer. Its main limitations are that probe structure remains visible, mixture ordering weakens somewhat, and the method is still a compact sklearn baseline rather than a dedicated invariant deep representation.